1 Features list

Categorized Features

This section shows the final list of features that have been used for predicting burned area and fire occurrence.

```
id_encoder
rhum_mean
Past_risk
fwi_max
cluster_encoder
isi_mean
rhum_max
bui_max
ffmc_mean
ffmc_max
dayofyear
sum_rain_last_7_days_mean
nesterov_mean
NDSI_min
calendar_sum
calendar_min
prec24h_max
angstroem_mean
sum_rain_last_7_days_max
munger_mean
dc_max
month
calendar_max
nesterov_max
rhum16_mean
precipitationIndexN5_max
precipitationIndexN5_mean
munger_max
angstroem_max
Pin maritime_max
precipitationIndexN3_max
foret_encoder_mean
Past_burnedarea
rhum_min
days_since_rain_max
temp_max
rhum16_max
Road_mean
```

temp16_max
fwi_min
NDSI_mean
sum_rain_last_7_days_min
prec24h16_max
dailySeverityRating_min
ffmc_min
bui_min
days_since_rain_min
Corine_vegetation_mean
rhum16_min
foret_encoder_max
Corine_vegetation_max
Chênes sempervirents_max
holidays
prcp_max
wdir16_mean
precipitationIndexN5_min
NDMI_min
munger_min
nesterov_min
Corine_grass_mean
wspd16_min
angstroem_min
Corine_urban_max
Mélèze_max
wdir_max
temp16_min
dc_min
dwpt_max
wdir_mean
population_mean
Pins mélangés_mean
Corine_littoral_max
Peuplier_max
dwpt16_max
NC_mean
NDMI_max
Mélèze_mean
Châtaignier_max
Pin laricio, pin noir_max
dwpt_min
NDSI_max
dwpt16_min
Corine_agricultural_mean
NDWI_max

Pins mélangés_max
wdir_min
NDMI_mean
Corine_water_mean
Pin d'Alep_max
Corine_grass_max
elevation_max
Corine_transport_mean
Conifères_mean
elevation_mean
wspd_max
NC_max
Pin autre_max
Peuplier_mean
prcp16_max
Douglas_mean
corine_encoder_mean
precipitationIndexN3_min
Feuillus_max
NR_max
Corine_rock_max
holidaysBorder
population_max
NDVI_mean
Pin à crochets, pin cembro_max
Sapin, épicéa_max
Conifères_max
NDVI_max
Pin sylvestre_mean
Mixte_max
Mixte_mean
Robinier_max
Pin laricio, pin noir_mean
wspd_mean
Douglas_max
wdir16_min
Corine_forest_mean
Pin sylvestre_max
Robinier_mean
wdir16_max
Chênes décidus_max
Corine_transport_max
Corine_water_max
Chênes décidus_mean
elevation_min
kbdi_max

```
wspd16_mean
Feuillus_mean
Corine_forest_max
Hêtre_mean
wspd_min
Hêtre_max
wspd16_max
prec24h16_min
prec24h_min
Corine_agricultural_max
corine_encoder_min
foret_encoder_min
ramadan
Corine_moisture_max
dayofweek
snow24h_max
snow24h16_max
sum_snow_last_7_days_max
isweekend
population_min
sum_snow_last_7_days_min
sum_consecutive_rainfall_max
prcp16_min
prcp_min
Road_min
snow24h_min
confinement
Feuillus_min
kbdi_min
sum_consecutive_rainfall_min
snow24h16_min
couvrefeux
```

2 Tree based configuration

Table 1 presents the configuration of tree based models.

3 Deep learning layers configuration

This section present the layers parameters for each deep learning model used in this study.

3.1 MLP

Table 2 shows the configuration of the MLP models.

Table 1: **Configuration of each tree based model**

Parameters	**XGBoost**	**Catboost**
Early Stopping Rounds	15	15
Learning Rate	0.001	0.001
Max Depth/Num Leaves	6	4
Min Child Weight	1.0	-
Max Delta Step	1.0	-
Subsample	0.5	0.7
Col Sample Bytree	0.7	0.6
Reg Lambda	1.7	1
Reg Alpha	0.7	0.27
N Estimators	10000	10000
Tree Method	hist	-

Table 2: Layer-wise architecture of the MLP model

Layer	**Configuration / Dimensions**
Linear 1	Linear(162 $\rightarrow$ 128), ReLU
Linear 2	Linear(162 $\rightarrow$ 256), ReLU
Linear 3	Linear(256 $\rightarrow$ 64), ReLU
Linear 4	Linear(64 $\rightarrow$ 5)

3.2 GraphCastGRU

Table 3 shows the configuration of the GraphCastGRU model. Edges channels is set to 4 as in the original paper containing the x, y, z positions of the mesh node and the length of the edge. Similarly, the input mesh data is the 3D position of the node. To better match with our spatial scale we used a mesh of order 4 instead of 6 in the original paper which offer better fits with our data.

Table 3: Layer-wise architecture of the `GraphCastGRU` model

Layer	**Configuration / Dimensions**
GRU	GRU(num_layers=2, input_size=`in_channels`, hidden_size=128)
BatchNorm	BatchNorm1d(128)
Dropout	Dropout(0.03)
GraphCastNet	Message Passing (6 layers), hidden dim = 128
Linear 1	Linear(128 $\rightarrow$ 256)
Linear 2	Linear(256 $\rightarrow$ 64)
Output Layer	Linear(64 $\rightarrow$ `out_channels`)
Activation (internal)	ReLU
Activation (output)	Softmax (for classification)

3.3 GRU

Table 4 shows the configuration of the GRU model.

Table 4: Layer-wise architecture of the GRU model

Layer	**Configuration / Dimensions**
GRU	2 layers, hidden size 128, ReLU, dropout 0.03
Norm 1	BatchNorm
Dropout	0.03
Linear 1	Linear(128 $\rightarrow$ 256), ReLU
Linear 2	Linear(256 $\rightarrow$ 64), ReLU
Linear 3	Linear(64 $\rightarrow$ 5)

3.4 LSTM

Table 5 shows the configuration of the LSTM model.

Table 5: Layer-wise architecture of the LSTM model

Layer	**Configuration / Dimensions**
LSTM	1 layer, hidden size 128, ReLU
Norm 1	BatchNorm
Dropout	0.03
Linear 1	Linear(128 $\rightarrow$ 256), ReLU
Linear 2	Linear(256 $\rightarrow$ 64), ReLU
Linear 3	Linear(64 $\rightarrow$ 5)

3.5 DilatedCNN

Table 6 shows the configuration of the DilatedCNN model.

Table 6: Layer-wise architecture of the DilatedCNN model (Dil = dilation)

Layer	**Configuration / Dimensions**
Conv 1	Conv1d(162 filters, dilation 1)
Conv 2	Conv1d(162 filters, dilation 3)
Conv 3	Conv1d(128, filters, dilation 4)
Norm + Dropout	BatchNorm, dropout 0.03
Linear 1	Linear(128 $\rightarrow$ 256)
Linear 2	Linear(256 $\rightarrow$ 64)
Linear 3	Linear(64 $\rightarrow$ 5)

4 Features importance

This section features importance based on shap values for Catboost models. Figure 1 shows features importance of Burned area prediction of Catboost model.

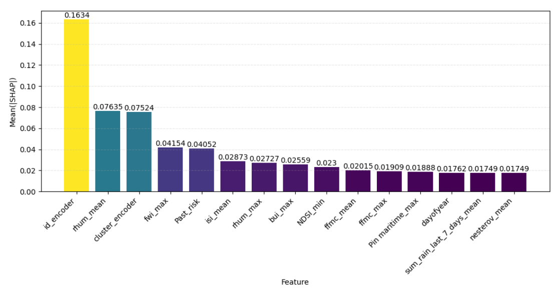

Figure 1: Top 15 shap values computed on multi-classification Catboost model for Burned area prediction. ID encoder correspond to Departmment ID